

HDOC Day-38

Write a menu-driven C program to perform the following operations using separate functions:

1. Count spaces, digits, and special characters in a string.
2. Count frequency of a given character in a string.
3. Toggle the case of each character in a string.
4. Exit.

1. Problem Understanding

The program reads a string and repeatedly displays a menu. According to the user's choice, it calls a separate function to count spaces, digits and special characters, count the frequency of a given character, or toggle the case of alphabetic characters. The program ends when the user selects Exit.

2. Standard Test Cases

Test Case	Menu Choice	Input String	Character Input	Expected Output
1	1	Hello 123!	-	Spaces = 1, Digits = 3, Special characters = 1
2	2	banana	a	Frequency of a = 3
3	3	HeLLo 123!	-	hElLO 123!
4	2	Programming	m	Frequency of m = 2
5	4	-	-	Program exits

3. Algorithm

Main Algorithm

1. Start.
2. Read a string from the user.
3. Display the menu.
4. Read the user's choice.
5. Use switch-case to call the required function.
6. Repeat the menu until choice 4 is selected.
7. Stop.

Function 1: Count spaces, digits and special characters

1. Initialize spaces, digits and special to 0.
2. Traverse the string character by character.
3. If the character is a space, increase spaces.
4. Else if it is between 0 and 9, increase digits.
5. Else if it is not an alphabetic character, increase special.
6. Display all three counts.

Function 2: Count frequency of a character

1. Read the character to be searched.
2. Initialize count to 0.
3. Traverse the string.
4. Whenever the current character equals the given character, increase count.
5. Display the frequency.

Function 3: Toggle case

1. Traverse the string.
2. If a character is lowercase, convert it to uppercase.
3. Else if a character is uppercase, convert it to lowercase.
4. Leave digits, spaces and special characters unchanged.
5. Display the toggled string.

4. C Program

```
#include <stdio.h>

void countCharacters(char str[])
{
    int i, spaces = 0, digits = 0, special = 0;
    for(i = 0; str[i] != '\0'; i++)
    {
        if(str[i] == ' ')
        {
            spaces++;
        }
        else if(str[i] >= '0' && str[i] <= '9')
        {
            digits++;
        }
        else if(!((str[i] >= 'A' && str[i] <= 'Z') || (str[i] >= 'a' && str[i] <= 'z'))))
        {
            special++;
        }
    }
    printf("Spaces = %d\n", spaces);
    printf("Digits = %d\n", digits);
    printf("Special characters = %d\n", special);
}

void countFrequency(char str[])
{
    int i, count = 0;
    char ch;
    printf("Enter the character: ");
    scanf(" %c", &ch);
    for(i = 0; str[i] != '\0'; i++)
    {
        if(str[i] == ch)
        {
            count++;
        }
    }
}
```

```

    printf("Frequency of %c = %d\n", ch, count);
}
void toggleCase(char str[])
{
    int i;
    for(i = 0; str[i] != '\0'; i++)
    {
        if(str[i] >= 'a' && str[i] <= 'z')
        {
            str[i] = str[i] - 32;
        }
        else if(str[i] >= 'A' && str[i] <= 'Z')
        {
            str[i] = str[i] + 32;
        }
    }
    printf("Toggled string = %s\n", str);
}
int main()
{
    char str[100];
    int choice;
    printf("Enter a string: ");
    fgets(str, 100, stdin);
    int i = 0;
    while(str[i] != '\0')
    {
        if(str[i] == '\n')
        {
            str[i] = '\0';
            break;
        }
        i++;
    }
    do
    {
        printf("\n1. Count spaces, digits and special characters\n");

```

```
printf("2. Count frequency of a character\n");
printf("3. Toggle case\n");
printf("4. Exit\n");
printf("Enter your choice: ");
scanf("%d", &choice);
switch(choice)
{
    case 1:
        countCharacters(str);
        break;
    case 2:
        countFrequency(str);
        break;
    case 3:
        toggleCase(str);
        break;
    case 4:
        printf("Program ended.\n");
        break;
    default:
        printf("Invalid choice.\n");
}
}
while(choice != 4);
return 0;
}
```

5. Compilation and Execution Evidence

```
kali@kali: ~/Desktop
Session Actions Edit View Help
(kali@kali)-[~/Desktop]
$ gedit question.c
** (gedit:7231): WARNING **: 14:30:11.676: Could not load Peas repository: Typelib file for namespace 'Peas', version '1.0' not found
** (gedit:7231): WARNING **: 14:30:11.676: Could not load PeasGtk repository: Typelib file for namespace 'PeasGtk', version '1.0' not found
(kali@kali)-[~/Desktop]
$ gcc question.c -o question
(kali@kali)-[~/Desktop]
$ ./question
Enter a string: Hello 123!
1. Count spaces, digits and special characters
2. Count frequency of a character
3. Toggle case
4. Exit
Enter your choice: 1
Spaces = 1
Digits = 3
Special characters = 1
1. Count spaces, digits and special characters
2. Count frequency of a character
3. Toggle case
4. Exit
Enter your choice: 3
Toggled string = hELLO 123!
1. Count spaces, digits and special characters
2. Count frequency of a character
3. Toggle case
4. Exit
Enter your choice: 4
Program ended.
(kali@kali)-[~/Desktop]
$ ./question
Enter a string: banana
```

```
kali@kali: ~/Desktop
Session Actions Edit View Help
(kali@kali)-[~/Desktop]
$ ./question
Enter a string: banana
1. Count spaces, digits and special characters
2. Count frequency of a character
3. Toggle case
4. Exit
Enter your choice: 2
Enter the character: a
Frequency of a = 3
1. Count spaces, digits and special characters
2. Count frequency of a character
3. Toggle case
4. Exit
Enter your choice: 4
Program ended.
(kali@kali)-[~/Desktop]
$ ./question
Enter a string: Programming
1. Count spaces, digits and special characters
2. Count frequency of a character
3. Toggle case
4. Exit
Enter your choice: 2
Enter the character: m
Frequency of m = 2
1. Count spaces, digits and special characters
2. Count frequency of a character
3. Toggle case
4. Exit
Enter your choice: 4
Program ended.
(kali@kali)-[~/Desktop]
$
```

6. Verification of All Test Cases

Test Case	Choice	String Input	Character	Actual Result	Status
1	1	Hello 123!	-	Spaces=1, Digits=3, Special=1	Pass
2	2	banana	a	Frequency=3	Pass

3	3	HeLLo 123!	-	hElLO 123!	Pass
4	2	Programming	m	Frequency=2	Pass
5	4	-	-	Program exits	Pass